

Teacher Guide — senior / module-03- ayurveda-good-health

IKS Curriculum

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Overview

Module 3 — Ayurveda: Classical Foundations and Modern Evidence

Age band: Senior · **Grades:** 9–12 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Locate** Ayurveda within the IKS map (Upaveda of the Atharva Veda) and identify the 8 branches (*aṣṭāṅga*) and 3 foundational texts (Caraka Saṃhitā, Suśruta Saṃhitā, Aṣṭāṅga Hṛdaya).
2. **Read** at least one primary source verse (Caraka Saṃhitā Sūtra-sthāna excerpt) and one BG verse (6.17) in IAST + translation.
3. **Articulate** the Tridosha theory at scholarly depth, including its dual nature (typology + medical framework) and contemporary scholarly engagement with it.
4. **Distinguish** the parts of Ayurveda that have strong modern evidence (turmeric anti-inflammatory action, time-restricted eating, sleep hygiene) from the parts that don't (e.g. dosha-as-strict-biomedical-classification).
5. **Engage** the contemporary debate about integrating Ayurveda into modern healthcare (AYUSH ministry, NHS pilot programmes, AIIMS integration units), articulating multiple defensible positions.
6. **Construct** a 5–6 page research-synthesis paper on a specific Ayurvedic claim, integrating classical primary sources, modern empirical research, and one historical/scholarly source.

NEP 2020 alignment

- §4.6 — IKS integration
- §4.31 — Physical fitness, well-being
- §11.8 — Multidisciplinary linkage

Prerequisites

- Module 2 (Panchabhuta) recommended — Tridosha builds on the five elements.
- Basic biology of digestion, circadian rhythms.
- Familiarity with structured argumentative writing + IAST reading.

Key vocabulary (Senior additions)

- *aṣṭāṅga Ayurveda* — the 8 branches
- *kāya-cikitsā* — internal medicine
- *śalya-tantra* — surgery (Sushruta's domain)
- *prakṛti* / *vikṛti* — constitution / current state
- *pramāṇa* — means of valid knowledge (epistemological framework)

Materials checklist

- Caraka Saṃhitā Sūtra-sthāna ch. 1 + ch. 26 (selected verses, IAST + translation)
- Patwardhan et al. (2015) extract on integrative approaches
- Sample modern RCT abstract on turmeric (Aggarwal & Harikumar 2009 or similar)
- BG 6.17 handout

Day-by-day overview

Day	Title	Focus
1	Ayurveda's place in IKS	8 branches + 3 foundational texts
2	Caraka Saṃhitā primary source	Close reading
3	Tridosha theory in depth	Classical + modern critique
4	Modern evidence base	What survives RCT, what doesn't
5	Mid-module + paper topic	Formative + outline
6	BG 6.17 + lifestyle medicine	Comparative framing
7	Integration debate	AYUSH, AIIMS, NHS pilots
8	Project Session 1	Outline + counter
9	Project Session 2	Draft
10	Final + paper	Summative + paper

Citation sources

See `sources.md`. Required for paper: ≥ 1 classical primary (Caraka, Suśruta, BG) + ≥ 1 modern empirical (peer-reviewed) + ≥ 1 historical/scholarly (Wujastyk, Patwardhan).

Project brief

5–6 page research-synthesis paper. See `assessments/project-brief.md`.

10-Day Lesson Plan

Module 3 (Senior) — 10-Day Lesson Plan

Day	Title	File
1	Ayurveda's place in IKS	days/day-01-hook.md
2	Caraka primary source	days/day-02-caraka.md
3	Tridosha in depth	days/day-03-tridosha.md
4	Modern evidence	days/day-04-modern-evidence.md
5	Mid + paper topic	days/day-05-mid-module.md
6	BG 6.17 + lifestyle	days/day-06-bg-lifestyle.md
7	Integration debate	days/day-07-integration-debate.md
8	Project Session 1	days/day-08-project-session-1.md
9	Project Session 2	days/day-09-project-session-2.md
10	Final + paper	days/day-10-final-assessment.md

Pacing

- Day 2 (Caraka reading) needs quiet time.
- Day 4 (modern evidence) requires students to engage RCT abstracts — non-trivial.
- Day 7 debate is loaded; honour the discomfort.

What students leave with

- Ayurveda mapped within IKS at scholarly depth.
- Primary source reading skill (Caraka in IAST).
- A defensible position on Ayurveda-in-modern-healthcare.
- One research paper integrating classical + empirical + historical sources.

Teacher Notes

Teacher Notes — Module 3 (Senior)

Posture

This is a serious comparative-medicine module. The honest framing:

Ayurveda is a sophisticated 2000-year-old framework for lifestyle medicine, daily wellbeing, and chronic care. Modern evidence supports much of it (turmeric, sleep timing, time-restricted eating). The dosha-as-strict-biomedical-classification has mixed evidence and shouldn't be oversold.

Three failure modes:

1. **Triumphalism** — “Ayurveda has cured cancer for 5000 years.” Specific claims, please, with sources.
2. **Dismissal** — “It’s all placebo.” Demonstrably false for several interventions.
3. **Boosterism without evidence** — “Ayurveda is just good.” This is hagiography, not scholarship.

Hard rule — medical caveat

Always: **nothing in this module is medical advice.** Students with medical conditions continue their existing care. If anyone seems to be planning to drop a treatment because of this module, intervene immediately.

The 8 branches (aṣṭāṅga Ayurveda)

1. *Kāya-cikitsā* — internal medicine
2. *Bāla-cikitsā* — pediatrics
3. *Graha-cikitsā* — psychiatry (literally “afflictions”)
4. *Ūrdhvāṅga-cikitsā* — ENT, ophthalmology
5. *Śalya-tantra* — surgery
6. *Daṁṣṭra-cikitsā* — toxicology
7. *Jarā-cikitsā* — geriatrics, rejuvenation
8. *Vṛṣa-cikitsā* — reproductive/aphrodisiacs

Foundational texts (the br̥hatrayī — “great three”)

- **Caraka Saṃhitā** (~2nd c. BCE) — internal medicine focus
- **Suśruta Saṃhitā** (~6th c. BCE or earlier?) — surgery focus
- **Aṣṭāṅga Hṛdaya** by Vāgbhaṭa (~7th c. CE) — synthesis, standard textbook

Common misconceptions

1. **“Ayurveda has no evidence base.”** False. Modest but real. Turmeric (curcumin) anti-inflammatory action — well-replicated. Yoga + breath for HRV — replicated. Tulsi adaptogenic effects — modest evidence. The picture is mixed but not empty.
2. **“Ayurveda has full evidence base.”** Also false. Many specific dosha-based prescriptions don’t hold up to RCT scrutiny. Dosha typology has weak biomarker correlates.
3. **“Pulse diagnosis (nāḍī parīkṣā) is scientifically validated.”** Mixed evidence. Skilled practitioners show some inter-rater agreement; biomarker validation is weak.

On the integration debate (Day 7)

Three positions in current scholarly + policy literature:

- **Full integration:** Ayurveda alongside biomedicine; AYUSH ministry approach in India.
- **Selective integration:** Use Ayurvedic interventions where evidence supports; reject where it doesn’t. (Patwardhan et al. approach.)
- **Separation:** Ayurveda as cultural-historical heritage; biomedicine for actual treatment. (Some Indian scientists and medical bodies, e.g. Indian Medical Association on certain points.)

All three are defensible. The skill is articulating each fairly.

Sensitivity

This is politically loaded in India. The AYUSH ministry is a real government body. Cultural pride is real. Skepticism is real. Honour both.

Project oversight

- Strong topics: “Turmeric (curcumin) — what does the modern evidence say about its anti-inflammatory claims?” or “Is the Tridosha typology validated by modern biomarker studies?”
- Weak topics: “Ayurveda is great” / “Ayurveda doesn’t work.”

If you only have 6 days

- Day 1: Hook + 8 branches
- Day 2: Caraka primary source
- Day 3: Tridosha + modern evidence
- Day 4: Integration debate
- Day 5: Project session
- Day 6: Final + paper

Sources

Sources — Module 3 (Senior)

Classical primary sources

- **Caraka Saṃhitā** (~2nd c. BCE) — *Sūtra-sthāna* chapters 1, 26 most relevant. Translation: Sharma, P. V. (1981). Chaukhambha Orientalia.
- **Suśruta Saṃhitā** — surgical procedures. Translation: Bhishagratna, K. K. (1907–1916), reprinted Chaukhambha.
- **Aṣṭāṅga Hṛdaya** by Vāgbhaṭa (~7th c. CE) — translation: Murthy, K. R. Srikantha (1991–95).
- **Bhagavad-gītā 6.17** — *yuktāhāra-vihārasya* — regulated lifestyle. Multiple translations.

Modern scholarly sources

- **Wujastyk, D. (2003).** *The Roots of Ayurveda*. Penguin. — Accessible primary-source translations.
- **Wujastyk, D., & Smith, F. M., eds. (2008).** *Modern and Global Ayurveda: Pluralism and Paradigms*. SUNY Press. — Engages the modern global spread.
- **Patwardhan, B., Mutalik, G., & Tillu, G. (2015).** *Integrative Approaches for Health: Biomedical Research, Ayurveda and Yoga*. Academic Press. — Engages the modern evidence question seriously.

Modern empirical research

- **Aggarwal, B. B., & Harikumar, K. B. (2009).** “Potential therapeutic effects of curcumin, the anti-inflammatory agent, against neurodegenerative, cardiovascular, pulmonary, metabolic, autoimmune and neoplastic diseases.” *International Journal of Biochemistry & Cell Biology*, 41(1), 40–59. — Strong review of curcumin evidence.
- **Patwardhan, B. (2014).** “Bridging Ayurveda with evidence-based scientific approaches in medicine.” *EPMA Journal*, 5(1), 19.
- **Panda, S. (2018).** *The Circadian Code*. Rodale. — Modern lifestyle medicine; aligns with dinacharya.
- **Streeter, C. C., et al. (2010).** “Effects of yoga versus walking on mood, anxiety, and brain GABA levels.” — Yoga as adjunct intervention.

Critical / policy sources

- **AYUSH Ministry, Government of India.** Various reports on Ayurvedic integration.
- **Indian Medical Association** statements on cross-pathy.
- **The Lancet** (occasional) on traditional medicine integration concerns.

RAG cache

. rag-cache/ — BG 6.17 + general Ayurveda framing chunks (copied from Middle band).

Citation discipline

A strong Senior paper cites at least one classical primary, one modern empirical, and one historical/scholarly source. Each cited substantively, not just name-checked. Invented citations score 0 on multiple criteria.

Day 1 — Ayurveda’s Place in IKS

Module: Ayurveda (Senior) · **Time:** 45 min

Learning objective

Students locate Ayurveda within the IKS map and name the 8 branches and 3 foundational texts.

Materials

- Whiteboard
- Reference card with the 8 branches

Flow

Warm-up (5 min)

Provocation: “*In what way is Ayurveda ‘science’? In what way is it not?*”

Take 3 student answers.

Core (25 min)

1. **Ayurveda’s place.** Ayurveda is an **Upaveda** — a “subordinate Veda.” Specifically, it’s traditionally an Upaveda of the **Atharva Veda** (which contained early medical material).
2. **The 8 branches (aṣṭāṅga Ayurveda):**

#	Sanskrit	Domain
1	Kāya-cikitsā	Internal medicine
2	Bāla-cikitsā	Pediatrics
3	Graha-cikitsā	“Psychiatry” / affliction medicine
4	Ūrdhvāṅga-cikitsā	ENT, ophthalmology
5	Śalya-tantra	Surgery
6	Daṁṣṭra-cikitsā	Toxicology
7	Jarā-cikitsā	Geriatrics, rejuvenation
8	Vṛ̥ṣa-cikitsā	Reproductive medicine

Note: this is a *comprehensive* medical framework, not just herbalism.

3. **The bṛhatrayī (“great three”) foundational texts:**

- **Caraka Saṃhitā** (~2nd c. BCE) — internal medicine focus. Attributed to Caraka; redacted by Dṛḍhabala.
 - **Suśruta Saṃhitā** (~6th c. BCE or later) — surgery focus. Suśruta is the source of the cataract-surgery and rhinoplasty procedures Module 1 covered.
 - **Aṣṭāṅga Hṛdaya** (~7th c. CE) — synthesis by Vāgbhaṭa; standard classical textbook.
4. **The honest scope:** Ayurveda was (and is) a comprehensive medical system covering internal medicine, surgery, pediatrics, psychiatry, ENT, toxicology. It is NOT just “herbs.”

Discussion (10 min)

- Which branch surprised you most?
- Where does Ayurveda overlap with modern medicine cleanly?

Wrap-up (5 min)

- Preview Day 2: “*Tomorrow — we read Caraka Saṃhitā Sūtra-sthāna ch. 1 in IAST.*”

Differentiation

- **Tier up:** Read Wujastyk (2003) intro chapter.
- **Tier down:** Memorise the 3 foundational texts + 4 of the 8 branches.

Teacher notes

- Many students think Ayurveda = herbal teas. The 8-branches list is a corrective.
- The Suśruta-surgery point ties to Module 1; reinforces continuity.

Citation(s) used

- Wujastyk (2003), introduction.
- Caraka Saṃhitā Sūtra-sthāna ch. 1 — Sharma (1981).

Day 2 — Caraka Saṃhitā: Primary Source Reading

Module: Ayurveda (Senior) · **Time:** 45 min

Learning objective

Students read selected verses from Caraka Saṃhitā Sūtra-sthāna in IAST + translation, identifying Caraka’s definition of *āyu* (life) and his epistemological framework.

Materials

- Printed handout: Caraka Sūtra-sthāna 1.41–45 (definition of *āyu*) in IAST + translation
- Optional: Caraka Sūtra-sthāna 11 (pramāṇa — means of valid knowledge)

Flow

Warm-up (5 min)

Recall: 3 foundational texts; 4 of the 8 branches.

Reading (25 min)

1. Caraka's definition of āyu (Sūtra-sthāna 1.42):

śarīrendryasattvātmasaṃyogo dhāri jīvitam nityagaścānubandhaśca paryāyair āyur ucyate

“The conjunction of body, senses, mind, and self — which sustains [life] — is called *jīvita* (life). Persistent and connected — these are synonyms for *āyu*.”

Word-by-word:

- *śarīra* = body
- *indrya* = senses
- *sattva* = mind
- *ātma* = self
- *saṃyogo* = conjunction, union
- *dhāri* = sustainer
- *jīvitam* = life

Note the move: āyu = body + senses + mind + self. **A four-component definition of life.** Not just biology — includes psychology and the witnessing self.

2. Caraka's epistemology (Sūtra-sthāna 11). Three pramāṇas (means of valid knowledge):

- *Āpta-upadeśa* — authoritative teaching (testimony)
- *Pratyakṣa* — direct perception
- *Anumāna* — inference

Note: this is essentially the same epistemological framework that the Nyāya school formalised (with one additional pramāṇa — comparison). Caraka was working within an articulated philosophy of knowledge.

3. Discussion:

- How does Caraka's 4-component āyu compare with the modern biomedical view of “life”?
- The pramāṇa framework predates Aristotle's *Posterior Analytics* in important ways. Compare.

Activity (10 min)

Quiet reading. Students annotate the verse handout. Note one question.

Wrap-up (5 min)

- Preview Day 3: “Tomorrow — Tridosha theory in classical depth + modern engagement.”

Differentiation

- **Tier up:** Read Sūtra-sthāna ch. 11 in full (Sharma 1981 translation).
- **Tier down:** Focus on the *āyu* definition; the *pramāṇa* material is bonus.

Teacher notes

- Caraka’s text is dense. Don’t try to cover everything. The *āyu* definition + the *pramāṇa* naming are enough.
- The “mind + self” inclusion in life surprises students raised on a body-only definition. Honour the surprise.

Citation(s) used

- Caraka Sūtra-sthāna 1.42 — Sharma (1981).
- vidya-karana RAG day-01-intro.json for supporting Bhagavad-gītā references.

Day 3 — Tridosha Theory in Depth + Modern Engagement

Module: Ayurveda (Senior) · **Time:** 45 min

Learning objective

Students articulate the Tridosha framework at classical depth + engage with the modern scholarly literature on its biomedical validity.

Materials

- Tridosha reference table
- Printed extract: brief abstract of a study attempting biomarker validation of dosha types (e.g. Bhalerao et al. or similar)

Flow

Warm-up (5 min)

Recall: 3 doshas with their elemental basis (built from Module 2 Panchabhuta).

Core (25 min)

1. Classical Tridosha (depth):

Dosha	Elements (mahābhūta)	Five sub-types	Primary site	Function
Vāta	ākāśa + vāyu	Prāṇa, Udāna,	Colon	Movement

Dosha	Elements (mahābhūta)	Five sub-types	Primary site	Function
		Samāna, Vyāna, Apāna	(pakvāśaya)	(nerve impulses, breath)
Pitta	tejas + ap	Pācaka, Rañjaka, Sādhaka, Ālocaka, Bhrājaka	Small intestine	Transformation (digestion, vision)
Kapha	ap + pṛthvī	Avalambaka, Kledaka, Bodhaka, Tarpaka, Ślaiṣmaka	Stomach (āmāśaya)	Structure, lubrication, immunity

Note: each dosha has FIVE sub-types with specific anatomical locations. The framework is far more granular than “Vāta = wind.”

2. Prakṛti vs Vikṛti:

- *Prakṛti* = constitution (set at birth).
- *Vikṛti* = current state (can be out of balance).
- Ayurveda diagnoses vikṛti, prescribes return-to-prakṛti.

3. The modern scholarly engagement:

- **Prasher et al. (2008).** “Genome-wide expression analysis of Prakriti types.” Compared gene expression in self-identified Vāta / Pitta / Kapha types. Found some differences. Replication mixed.
- **Patwardhan (2014).** *Bridging Ayurveda with evidence-based scientific approaches.* Argues for “Ayurgenomics” — combining Ayurvedic typology with genomic and metabolomic data. Promising; not conclusive.
- **Critical view:** even the modest published correlations may be confounded by self-identification bias. Dosha typology may correlate with personality traits more than with biomedical biomarkers.

4. **The honest framing:** the Tridosha framework is *useful* clinically for many practitioners. The *biomedical-classification* claim has mixed evidence. These are different claims. Don’t conflate.

Discussion (10 min)

- “Is Tridosha ‘true’ in the same way the periodic table is true? In what way might it be a different kind of framework?”
- Push toward: clinical typologies (DSM personality types, Sheldon’s somatotypes) often have modest biomarker correlates but real clinical utility. Tridosha may be in this category.

Wrap-up (5 min)

- Preview Day 4: “Tomorrow — the broader modern evidence base for Ayurveda.”

Differentiation

- **Tier up:** Read the Prasher et al. (2008) abstract carefully. Critique the methodology.
- **Tier down:** Focus on classical Tridosha + the framing distinction (clinical vs biomedical).

Teacher notes

- The Tridosha “is it true?” question doesn’t have a clean yes/no. Stay with the framing distinction.
- Some students will want Tridosha to be either ALL true or ALL false. Both positions are too crude.

Citation(s) used

- Caraka Sūtra-sthāna 18 on dosha sub-types.
- Patwardhan (2014). *EPMA Journal* 5(1), 19.
- Prasher et al. (2008) on Prakriti genomics (debated).

Day 4 — Modern Evidence Base for Ayurveda

Module: Ayurveda (Senior) · **Time:** 45 min

Learning objective

Students review the modern evidence base for Ayurveda, distinguishing well-evidenced interventions from poorly-evidenced ones.

Materials

- Printed abstract: Aggarwal & Harikumar (2009) on curcumin
- Printed extract: Panda (2018) on time-restricted eating
- Whiteboard

Flow

Warm-up (5 min)

Recall: Tridosha clinical vs biomedical framing distinction.

Core (25 min)

1. **Build the evidence map** on the board:

Intervention	Modern evidence	Quality
Turmeric (curcumin) — anti-inflammatory	Well-replicated; reviews in <i>Int J Biochem Cell Biol</i>	Strong
Time-restricted eating (aligned with dinacharya)	Panda et al. — strong evidence for metabolic benefits	Strong

Intervention	Modern evidence	Quality
midday-meal principle)		
Yoga + breath for HRV	Streeter (2010), multiple studies	Moderate
Tulsi (Ocimum sanctum) adaptogenic	Modest RCT support for stress reduction	Modest
Triphala for digestive function	Some clinical evidence	Modest
Pulse diagnosis (nāḍī parīkṣā)	Inter-rater agreement studies — mixed	Weak
Specific dosha-based dietary prescriptions	Mixed; many specific claims fail RCT	Weak
Pancha-karma cleansing protocols	Limited; mostly observational	Weak

2. **The pattern.** Things that survive evidence scrutiny:

- Single-compound herbs with measurable biochemical effects (turmeric, ashwagandha, tulsi).
- Lifestyle / behavioral interventions aligned with modern lifestyle medicine (regular sleep, meal timing, exercise).
- General yoga and breath interventions.

Things that don't survive as well:

- Specific dosha-type prescriptions for unrelated conditions.
- Mechanism claims that contradict modern physiology.

3. **The honest framing:** “Ayurveda has *some* strong evidence and *some* weak evidence. The mature stance is to use the strong, study the moderate, and be skeptical of the weak.”

Activity (10 min) — Read a real abstract

Hand out the Aggarwal & Harikumar (2009) curcumin review abstract. Students read silently. Then answer: - What conditions does it discuss curcumin for? - What's the strength of the evidence per condition? - What hedge words does the abstract use?

Wrap-up (5 min)

- Preview Day 5: “*Tomorrow — mid-module + paper topic commitment.*”

Differentiation

- **Tier up:** Read a full RCT — find one via Google Scholar on a specific Ayurvedic intervention. Critique.
- **Tier down:** Memorise the evidence-strength table.

Teacher notes

- The “evidence map” is the most important takeaway. Distinguish well-evidenced from weakly-evidenced. Don't lump.

- Some students will resist the “weak evidence” verdict on parts they personally value. Validate the value; hold the evidence standard.

Citation(s) used

- Aggarwal, B. B., & Harikumar, K. B. (2009). *Int J Biochem Cell Biol*, 41(1), 40–59.
- Panda, S. (2018). *The Circadian Code*. Rodale.
- Streeter, C. C., et al. (2010). *J Altern Complement Med*, 16(11), 1145–1152.

Day 5 — Mid-Module + Paper Topic

Module: Ayurveda (Senior) · **Time:** 45 min

Flow

Quiz (20 min)

Formative Part B, 12 marks.

Topic commitment (20 min)

Each student writes: - Working title - One-paragraph thesis (specific Ayurvedic claim being examined) - 3 sources: ≥1 classical primary, ≥1 modern empirical, ≥1 historical/scholarly

Suggested topics: - “Curcumin’s anti-inflammatory action — what does the modern evidence say?” - “The Tridosha typology: clinical heuristic or biomedical classification?” - “Dinacharya and circadian biology — convergent or coincidental?” - “Should AYUSH practitioners be allowed to prescribe biomedical drugs?” - “Sushruta’s surgical legacy — what specifically did European surgeons learn?”

Wrap-up (5 min)

- Topic + thesis submitted at door.

Day 6 — BG 6.17 + Lifestyle Medicine Convergence

Module: Ayurveda (Senior) · **Time:** 45 min

Learning objective

Students read BG 6.17 in IAST + translation and articulate the convergence between traditional dinacharya and modern lifestyle medicine.

Materials

- BG 6.17 handout (IAST + translation + Bryant commentary)
- Panda (2018) extract on time-restricted eating

Flow

Warm-up (5 min)

Recall: 5 dinacharya practices.

Reading (15 min)

yuktāhāra-vihārasya yukta-ceṣṭasya karmasu yukta-svapnāvabodhasya yogo bhavati duḥkha-hā — Bhagavad-gītā 6.17

Word-by-word + translation (as in Middle Day 7 — see also [curriculum/middle/module-03-ayurveda-good-health/days/day-07-primary-source.md](#)).

The convergence (15 min)

Build comparison table:

Traditional principle (Caraka, BG)	Modern lifestyle medicine equivalent	Evidence
Wake before sunrise	Aligning with circadian rhythm	Nobel 2017 — circadian biology
Largest meal midday (Pitta peak)	Time-restricted eating; metabolic window	Panda et al. (2018)
Sleep by ~10pm	Aligning with melatonin curve	Sleep science (Walker 2017)
Regular exercise to 50% capacity	Moderate-intensity training	Standard exercise physiology
Six tastes per meal	Macronutrient diversity, polyphenol intake	Nutrition science

Note the convergence is striking but not identical. The tradition observed; modern science measures. They overlap in many places; they're not the same project.

Discussion (8 min)

- “Is the convergence evidence that the tradition was ‘right’? Or that human physiology hasn’t changed?”
- Both. Don’t force a binary.

Wrap-up (2 min)

- Preview Day 7: “Tomorrow — the contemporary integration debate.”

Differentiation

- **Tier up:** Read Walker (2017) *Why We Sleep* chapter on circadian.
- **Tier down:** Focus on the convergence table.

Teacher notes

- Convergence is a powerful pedagogical moment but DON’T oversell (“Ayurveda predicted everything”).
- The honest framing: tradition + science arrived at overlapping conclusions by different methods.

Citation(s) used

- BG 6.17 — vidya-karana RAG cache.
- Bryant (2009).
- Panda (2018), Walker (2017).

Day 7 — The Integration Debate

Module: Ayurveda (Senior) · **Time:** 45 min

Learning objective

Students articulate three positions on integrating Ayurveda into modern healthcare and defend one with steel-manned counter-arguments.

Materials

- Whiteboard
- Optional: brief printed extracts on AYUSH ministry policy + Indian Medical Association statements

Flow

Warm-up (3 min)

Brief: today is structured debate. Three positions.

Three positions (5 min)

Position	One-line	Defenders
A — Full integration	Ayurveda alongside biomedicine; AYUSH practitioners can offer first-line care	AYUSH Ministry; many Indian Ayurveda practitioners
B — Selective integration	Use Ayurveda where evidence supports; biomedicine where it leads	Patwardhan et al. (2015); evidence-based traditional medicine movement
C — Separation	Ayurveda as cultural-historical study; biomedicine for actual treatment	Indian Medical Association (some positions); skeptical scientists

Debate (25 min)

- 5 min: groups by position.
- 10 min: prep — 2 arguments + 1 steel-manned counter to each.
- 10 min: structured debate (3 min/group + 1 min cross).

Oral assessment (during)

Teacher scores 5 random students on: position clarity, evidence use, counter-argument fairness, listening.

Wrap-up (12 min)

- 3 students share if their view shifted.
- Note: *“This is the most politically loaded session. Both extreme positions have problems. The mature position is somewhere in the middle, defended with evidence.”*

Differentiation

- **Tier up:** Take position you disagree with.
- **Tier down:** Use prepared 2-min argument.

Teacher notes

- AYUSH ministry is real, contemporary, politically significant. Honour the political reality.
- Some students from medical families have strong priors. Help them articulate their position rigorously.

Citation(s) used

- AYUSH Ministry policy documents.
- Patwardhan et al. (2015).
- Various IMA statements.

Day 8 — Project Session 1: Outline + Counter

Module: Ayurveda (Senior) · **Time:** 45 min

Flow

Sprints (40 min)

- 10 min: thesis + outline refinement
- 15 min: buddy critique — strongest counter
- 15 min: draft introduction

Wrap-up (5 min)

- Outline submitted.

Homework

- Draft 2 pages tonight.

Day 9 — Project Session 2: Draft

Module: Ayurveda (Senior) · **Time:** 45 min

Flow

Sprints (40 min)

- 20 min: silent drafting
- 15 min: buddy structural feedback
- 5 min: revise

Wrap-up (5 min)

- Submit polished intro + 1 body section.

Homework

- Complete the paper.

Day 10 — Summative + Paper Submission

Module: Ayurveda (Senior) · **Time:** 45 min

Flow

Quiz (35 min)

35 marks.

Submission (10 min)

- Hand in paper. Spot-check.
- Reflection slip.

Homework

- None.

Module 3 (Senior) — Formative Quizzes

Part A — Day 3 (6 Q, 10 min)

1.
(1) Which Veda is Ayurveda traditionally an Upaveda of?
2.
(1) Name 3 of the 8 branches.
3.
(2) Name the 3 foundational texts (bṛhatrayī).
4.
(1) Caraka defines āyu (life) as a conjunction of which four components?
5.
(2) Name the 3 pramāṇas (means of valid knowledge) Caraka articulates.

Key

1. Atharva Veda. 2. Any 3 of: Kāya-cikitsā, Bāla-cikitsā, Graha-cikitsā, Ūrdhvāṅga-cikitsā, Śalya-tantra, Daṁṣṭra-cikitsā, Jarā-cikitsā, Vṛṣa-cikitsā. 3. Caraka Saṃhitā, Suśruta Saṃhitā, Aṣṭāṅga Hṛdaya (Vāgbhaṭa). 4. śarīra (body) + indriya (senses) + sattva (mind) + ātman (self). 5. Āpta-upadeśa, Pratyakṣa, Anumāna.

Part B — Day 5 (12 marks, 15 min)

1.
(2) Tridosha sub-types: name the 5 sub-types of any ONE dosha.
2.
(2) Prakṛti vs Vikṛti — distinguish.
3.
(2) Patwardhan (2014) coined what term to describe combining Ayurvedic typology with modern omics?
4.
(2) Name 3 Ayurvedic interventions with STRONG modern evidence.
5.
(2) Name 2 with WEAK evidence.
6.
(2) Briefly state the 3 positions on Ayurveda's integration into modern healthcare.

Key

1. (e.g. Vāta): Prāṇa, Udāna, Samāna, Vyāna, Apāna. 2. Prakṛti = inborn constitution; Vikṛti = current state. 3. “Ayurgenomics.” 4. e.g. turmeric/curcumin anti-inflammatory, time-restricted eating (dinacharya), yoga for HRV. 5. e.g. pulse diagnosis (nāḍī parīkṣā), specific dosha-based prescriptions for unrelated conditions. 6. (A) Full integration (AYUSH); (B) Selective integration (Patwardhan); (C) Separation (cultural-historical study, biomedicine for treatment).

Module 3 (Senior) — Summative Quiz

Time: 35 min · **Marks:** 35 · **Closed-book**

Name: _____ **Class:** _____

A — Multiple choice (5 × 1 = 5)

1. Ayurveda is traditionally an Upaveda of:
a) Ṛg Veda b) Yajur Veda c) Sāma Veda d) Atharva Veda
2. The br̥hatṭrayī (“great three”) texts of Ayurveda are:
a) Vedas + Upaniṣads + Bhagavad-gītā
b) Caraka + Suśruta + Aṣṭāṅga Hṛdaya
c) Caraka + Bhāgavatam + Sūrya-siddhānta
d) Sūtra-sthāna + Cikitsā-sthāna + Kalpa-sthāna
3. Caraka’s definition of āyu includes:
a) body only
b) body + breath
c) body + senses + mind + self
d) body + soul + karma + dharma
4. Which intervention has STRONG modern evidence?
a) Pulse diagnosis b) Dosha-based career advice
c) Turmeric anti-inflammatory d) Pancha-karma for cancer
5. Patwardhan’s “Ayurgenomics” refers to:
a) Genetic Ayurveda b) Combining Ayurvedic typology + omics
c) Ayurvedic clones d) Vedic genetics

B — Primary source (5)

6.
(3) Translate word-by-word: *śarīrendṛyasattvātmasaṁyogo dhāri jīvitam* (Caraka Sūtra 1.42).
7.
(2) BG 6.17 names 5 activities to regulate. Name them.

C — Tridosha (5)

8. Name the 5 sub-types of Pitta with their primary functions/locations.

D — Evidence base (5)

9.
 - (3) Name 2 Ayurvedic interventions with STRONG modern evidence and 1 with WEAK evidence; cite a study or scholar.
10.
 - (2) The Tridosha typology — clinical heuristic vs biomedical classification. Distinguish.

E — Integration debate (5)

11.
 - (3) Briefly state the 3 positions on integrating Ayurveda into modern healthcare.
12.
 - (2) Steel-man the position you DON'T hold.

F — Open argument (10)

13.
 - (10) In 1–2 pages, argue:
“The honest framing of Ayurveda for modern students is neither ‘all true’ nor ‘all placebo,’ but a discrimination between well-evidenced and weakly-evidenced claims.”

Required: clear thesis, ≥1 source, steel-manned counter.

Key (selected)

1d. 2b. 3c. 4c. 5b. 6. śarīra (body) + indriya (senses) + sattva (mind) + ātman (self) + saṁyogo (conjunction) + dhāri (sustainer) + jīvitam (life). 7. eating, recreation, work, sleep, waking. 11. (A) Full integration (AYUSH). (B) Selective integration (Patwardhan). (C) Separation.

Total: 35.

Homework

Homework — Module 3 (Senior)

Day	Task	Time
1 → 2	Read Wujastyk (2003) intro.	30 min
2 → 3	Re-read Caraka 1.42 verses; note 2 questions.	20 min
3 → 4	Look up Prasher et al. (2008) abstract. Read carefully.	30 min

Day	Task	Time
4 → 5	Submit paper plan: thesis + 3 sources.	30 min
5 → 6	Read Panda (2018) excerpt on circadian.	30 min
6 → 7	Choose debate position; prepare 3-min argument.	45 min
7 → 8	Draft 1 page of paper.	60 min
8 → 9	Continue drafting (2+ pages).	60 min
9 → 10	Finalise, proofread, format.	60 min

Writing Prompts

Writing Prompts — Module 3 (Senior)

1. **(Day 2)** Caraka's 4-component definition of āyu (body + senses + mind + self) compared with the modern biomedical definition of life. Defend or critique. 1 page.
2. **(Day 3)** Tridosha as clinical heuristic vs biomedical classification — defend the distinction in 1 page.
3. **(Day 4)** Read the Aggarwal & Harikumar (2009) curcumin abstract. Critique its methodology or scope in 1 page.
4. **(Day 6)** Dinacharya + circadian biology — same insight or convergent inquiry? 1 page.
5. **(Day 7, paper potential)** Argue your position on Ayurveda's integration into modern healthcare. Steel-man the other two. 2 pages.
6. **(Cross-disciplinary)** Compare Caraka's pramāṇa epistemology with Aristotle's *Posterior Analytics*. What's similar, what's different? 2 pages.
7. **(Stretch)** Read one Indian Medical Association statement on cross-pathology. Engage it. 2 pages.

Activity 01 — Read an RCT Abstract

Module: Ayurveda (Senior) · **Day:** 4 · **Time:** 15 min within Day 4

Goal

Students read a real peer-reviewed abstract on an Ayurvedic intervention and critique it.

Materials

- Printed abstract: Aggarwal & Harikumar (2009) on curcumin (or substitute another)

Instructions

Read silently. Then answer in your workbook:

1. What's the intervention?
2. What conditions does the abstract cover?
3. What strength of evidence does it claim?
4. What hedge words appear? ("may," "possibly," "limited data," "further research needed")
5. What would a strong skeptic critique?

Discussion (5 min)

Pair share answers.

Differentiation

- **Tier up:** Find a related full RCT via Google Scholar; critique sample size, controls, blinding.
- **Tier down:** Focus on Qs 1–3.

Project Brief

Project Brief — Module 3 (Senior)

Type: 5–6 page research-synthesis paper • **Marks:** 65

Topics

Choose one or propose:

1. **A specific intervention** — e.g. "Curcumin's anti-inflammatory action — what does the evidence say?" or "Triphala for digestive function — review."
2. **A specific framework claim** — e.g. "Is the Tridosha typology validated by modern biomarker studies?"
3. **A specific convergence** — e.g. "Dinacharya and circadian biology — same insight or convergent inquiry?"
4. **A specific policy question** — e.g. "Should AYUSH practitioners be allowed to prescribe biomedical drugs?"
5. **A specific historical question** — e.g. "Sushruta's surgical legacy — what specifically did European surgeons learn?"

Required elements

- Clear thesis in opening.

- ≥1 classical primary source (Caraka, Suśruta, Aṣṭāṅga Hṛdaya, BG) — IAST + translation.
- ≥1 peer-reviewed modern empirical study.
- ≥1 historical / scholarly source (Wujastyk, Patwardhan).
- Steel-manned counter-argument.
- MLA or Chicago citations.

Format

5–6 pages, double-spaced, 12pt.

Schedule

- Day 5: topic + thesis.
- Day 8: outline.
- Day 9: draft.
- Day 10: paper submitted.

Rubric

See `rubric.md`.

Self-reflection (1 paragraph)

1. Topic + why.
2. Hardest part.
3. With 3 more days, deeper research on.

Rubric

Paper Rubric — Module 3 (Senior)

Paper marks: 65 (+ summative 35 → 100 total).

Criterion	4	3	2	1	Weight
Thesis clarity	Crystal-clear, specific	Clear	Vague	None	× 3
Classical primary source	IAST + translation + interpreted	Cited + interpreted	Vague	None	× 2

Criterion	4	3	2	1	Weight
Modern empirical source	Peer-reviewed study cited + critiqued	Cited	Vague	None	× 2
Historical/scholarly source	Substantive engagement	Cited	Vague	None	× 2
Counter-argument	Steel-manned + reasoned response	Counter presented	Acknowledged	Straw-manned	× 2
Evidence-discrimination	Distinctions strong / moderate / weak claims	Mostly	Some	Lumps	× 1
Avoidance of false dichotomy	Neither “all true” nor “all placebo”	Mostly nuanced	Hedges	Reduces	× 1
Prose / structure	Polished	Clear	Issues	Hard to follow	× 1

Max raw: 56. Paper = (sum × weights) × 65/56.

Deductions

- Outside 5–6 pages: –5
- Missing works-cited: –5

- IAST diacritics absent where Sanskrit appears: -3
- Recommending dropping medical treatment based on the module: -10 + redo